

Introduction to Machine Learning 1 (for Everybody)

ID: MLW-001

About This Course

Target Group

Anybody who wants to understand what Machine Learning is about and how it could be useful in our business.

Content

Machine learning is one domain of artificial intelligence and the core of many impressive results Google, Facebook and others present to us every day. But what is Machine learning exactly? How does it work? And how can it actually help also our business? This training will give an introduction for beginners. On the end you will be familiar with some tools, there pros and cons of Machine Learning and will even have built a neuronal network yourself.

- What is Machine Learning?
- Machine Learning types
- Classification Algorithms
- Neuronal Network Regression
- Application of Machine Learning in our business
- Limits of Machine Learning (Bias and Overfitting)
- Pros and cons of machine learning

Prerequisites

- Laptop (if classroom training)
- EnergyAI Account (Go to: <https://energyai.siemens.cloud/> (using Chrome, Edge or Firefox; Zscaler has to be active) and Request access to "Analyse" at least 1 week before the start of the training.)

Introduction to Machine Learning 2 (Adding Background)

ID: MLW-002

About This Course

Target Group

Employees who are planning to actually use ML for their work and therefore need to understand some background to apply it properly.

Content

In this training we will build on the training Machine Learning 1 and dig deeper. We will explain some critical mathematical background which will make it easier for you to interpret the results of decision tree, random forest and neuronal networks. The goal is, that you can build simple models yourself and to have an introduction to some basic concepts. You will know enough to start further exploring and learning.

- Mathematical background on Decision Tree analysis
- Mathematical background on Random Forrest analysis
- Mathematical background on Neuronal Network Regression
- Limits of Machine Learning (Bias and Overfitting)
- Difference Multiple Regression vs. Neuronal Network Regression
- Engineering application examples
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Prerequisites

- Attending "Introduction to Machine Learning 1 (for Everybody), MLW-001" or read through slides
- Laptop (if classroom training)
- EnergyAI Account (Go to: <https://energyai.siemens.cloud/> (using Chrome, Edge or Firefox; Zscaler has to be active) and Request access to "Analyse" at least 1 week before the start of the training.)